

Product Requirements Document

Nova Trade Document Intelligence - Part 1

Owner: Full-Stack AI Engineer candidate

Pilot customer: Acme Retail India

Scope: One-document validation foundation; Part 2 email trigger and multi-document workflow are explicitly out of scope.

1. Nova, FDE, and the System of Outcomes

What is Nova?

Nova is an agentic operations layer for global trade. Traditional SaaS records shipments, exposes workflows, and waits for people to move work forward. Nova instead reads unstructured inputs, applies customer-specific operating rules, decides the next safe action, and completes the repetitive portion of the job. The target is not another dashboard; it is a measurable operational result: fewer manual checks, fewer document-error cycles, and faster clearance. Because trade documents are messy and rules vary by customer, Nova must combine models, deterministic controls, tools, durable state, and human review. The value comes from owning the path from input to verified outcome while making uncertainty, evidence, and accountability visible.

What is the FDE model, and why use it?

A Forward Deployed Engineer works close to the customer's operation, combining engineering with discovery and implementation. The FDE observes how CG operators actually validate documents, turns informal rules into explicit controls, connects Nova to the customer's systems, and measures whether the result improves. GoComet uses this model because trade operations are not standardized enough for a generic prompt or configuration screen. Each rollout has different documents, terminology, exceptions, integrations, risk tolerance, and approval policy. Tight deployment loops let the engineer ship a narrow outcome quickly, learn from real exceptions, and convert customer-specific insight into reusable platform capability.

What does "System of Outcomes" mean?

A System of Record stores authoritative facts; a System of Engagement helps people communicate around those facts. A System of Outcomes is accountable for moving work to a defined result. Here, the result is not "invoice uploaded" or "comment added." It is "document verified, exception explained, and the correct next action prepared." Nova consumes records and communication, but it also interprets, validates, routes, persists, and measures work. Human operators retain control over ambiguous or high-risk cases. The difference is completion with guardrails: the product is judged by cycle time and safe straight-through processing, not by seats, clicks, or records created.

2. Problem Statement

Today CG operators open emailed attachments, read every field, recall customer-specific rules, compare values, type discrepancies, and repeat this process after every amendment. The flow breaks through:

- illegible scans, inconsistent layouts, and missing fields;
- rules living in people's heads, spreadsheets, and email history;

- transcription errors and inconsistent normalization of names, ports, units, Incoterms, and HS codes;
- silent misses when an operator is rushed or inexperienced;
- two-to-four amendment cycles, each adding 4–24 hours;
- no shared view of pending, approved, and disputed documents;
- weak auditability: the source evidence and decision reason are not linked.

In the operator's first five minutes, success means they can upload a document, see every required field with confidence and source evidence, understand which rules passed or failed, and accept a safe next action without re-reading the whole document. Uncertainty must be more obvious than approval.

3. Users and Jobs-to-be-Done

CG operator: validates high document volumes under time pressure. Cares about finding exceptions quickly, avoiding costly misses, and producing a complete amendment request.

SU document coordinator: creates and resubmits shipment documents. Cares about receiving one precise correction list instead of vague or repeated feedback.

Jobs-to-be-Done

1. When a supplier document arrives, I want required fields extracted with source evidence, so that I do not re-read the entire file.
2. When a value violates a customer rule, I want found and expected values shown together, so that I can verify the discrepancy in seconds.
3. When the scan is ambiguous, I want the system to block silent approval and identify the uncertain field, so that I retain control of risk.
4. When several discrepancies exist, I want one editable amendment draft, so that the supplier receives complete feedback in one cycle.
5. When a shipment is cleared, I want the decision and evidence stored, so that I can answer audits without reconstructing email history.
6. When I manage the queue, I want grounded questions over verified runs, so that I can see backlog and exception trends without asking an engineer.

4. Agent Architecture

Three agents align with three different kinds of work and failure:

1. **Extractor (probabilistic perception):** input is a PDF/image; output is a typed document object containing value, confidence, source snippet, and page for each required field.
2. **Validator (policy verification):** input is the extraction plus a versioned customer rule set. An LLM handles semantic equivalence and returns a structured field result; a deterministic rule tool independently checks requiredness, confidence, and exact policy. Disagreement becomes uncertain.
3. **Router (bounded decision):** input is the complete validation. An LLM proposes the outcome, explanation, and editable draft; a deterministic guard constrains the final output to one of `auto_approve`, `human_review`, or `amendment_request`, with reasoning and an editable draft when relevant.

One giant prompt mixes perception, policy, and action, making errors hard to localize and tests hard to reproduce. Five agents would split cohesive decisions without adding a distinct trust boundary. Three lets the model do perception while

code owns policy and routing. Agents communicate through versionable JSON contracts, not free-form conversation. SQLite holds the run after each stage; after a crash the pipeline can resume from the last valid checkpoint by run ID and content hash.

5. LLM and Tooling Choices

- **Extraction:** Gemini 3.6 Flash is the primary free-tier vision model because it accepts PDFs natively and supports JSON Schema output. Groq Qwen 3.6 27B is the fallback for images or rendered PDF pages. Both providers are configurable through environment variables rather than embedded in policy.

A smaller model limits cost and latency; hard cases can escalate to a stronger model only after a low-confidence or schema failure.

- **Validation:** deterministic Python first enforces equality, requiredness, and confidence thresholds. The selected provider is called only when a mismatch in a textual field may be an alias, abbreviation, or semantically equivalent description. Any model/rule disagreement becomes uncertain.
- **Routing:** deterministic policy controls the three bounded outcomes. Clean auto-approval uses no model call; the text model is invoked only to explain an exception or draft an editable amendment request, and cannot override policy.
- **Orchestration:** LangGraph models Extractor, Validator, and Router as explicit nodes. A SQLite checkpointer saves typed state after every node using the run ID as the thread ID. This provides durable recovery, state inspection, and a clean path to Part 2 conditional routing and human-review interrupts.
- **Structured output:** required at the extractor boundary; validated again in code. Free-form model output is avoided for rules, decisions, database queries, and any approval control.
- **Bad documents:** use high-detail input, never infer absent text, retry at most once with a stronger model or rendered pages, then route low confidence to a human. The POC exposes this state rather than hiding it.

6. Trust, Failure Handling, and Evals

Every non-null extracted value must include a visible source snippet and page. Missing evidence lowers confidence below the approval threshold. Required fields, model/schema failures, and low confidence fail loud. A document cannot be auto-approved if any field is uncertain or mismatch. Network/model retries are bounded at two attempts, stages have timeouts, files are size/type checked, and validation/routing never call themselves.

Offline eval: a versioned set of at least 100 documents with field-level ground truth, including 30 degraded scans. Report exact match/F1 by field, evidence coverage, calibration error, false-approval rate, and route accuracy. The release gate is zero false approvals on the high-risk test set and at least 95% field accuracy on clean documents.

Online metrics: sample completed and human-reviewed runs; compare operator corrections against extracted values and recommended routes. Alert on false approval, confidence drift, rising uncertainty by supplier/layout, model latency, and cost.

7. Metrics and Success Criteria

North star: Percentage of submitted documents safely resolved without manual

field-by-field reading, with zero confirmed false approvals.

Supporting metrics:

- field exact-match rate, overall and by required field;
- high-risk false-approval rate;
- evidence coverage rate;
- confidence calibration error;
- percentage routed to human review;
- median upload-to-decision latency and p95 latency;
- average model cost per document;
- amendment cycles per shipment;
- operator handling minutes per document.

Two-week pilot Go: at least 100 documents processed; $\geq 60\%$ safe straight-through resolution; zero confirmed false approvals; $\geq 95\%$ clean-document field accuracy; median decision under 60 seconds; and $\geq 40\%$ lower operator handling time.

No-Go: any confirmed false approval on a high-risk required field, evidence coverage below 99%, availability below 95%, or operators report that reviewing the output takes as long as the current process. A No-Go means retain human review, analyze errors by layout/rule, and do not expand traffic.

8. What Comes Next

With two more weeks, build the trigger and multi-document shipment model: ingest a simulated SU inbox, group BOL/invoice/packing-list attachments, cross-check shared fields, and present an editable reply for CG approval. This is next because it closes the gap between an upload demo and the real operational outcome while reusing every Part 1 boundary. Autonomous sending, broad ERP integrations, and fine-tuning wait until real exception data proves where they add value.

Note: This PRD is grounded in the supplied DAW brief and cross-checked against GoComet's public Nova Full-Stack AI Engineer job description, including its governed-agent, LangGraph, evidence-delivery, and FDE outcome-ownership model.